PROBLEM STATEMENT

Patient Data Management System Using Doubly Linked List

A hospital or clinic needs to maintain patient records such as Patient ID, Name, Age, and Disease. Since patients may be added or removed at any time, a dynamic data structure is required to manage the records efficiently.

The problem is to design and implement a Patient Data Management System using a Doubly Linked List. Each node of the doubly linked list should store the details of one patient along with two pointers: Previous and Next, which connect the node to the preceding and succeeding patient records.

The system should allow the user to perform basic patient management operations such as adding a new patient, displaying patient records, searching for a patient, deleting a patient record, and displaying records in both forward and reverse directions.

The implementation should maintain the links correctly after every insertion and deletion operation and demonstrate the use of dynamic memory allocation and doubly linked list traversal.

Objective

To develop a simple and efficient patient record management system using a Doubly Linked List and understand dynamic memory allocation, insertion, deletion, searching, and bidirectional traversal.

Required Operations

1. Add Patient – Insert a new patient record.

2. Display Patients – Display all records in forward order.

3. Display in Reverse – Display records from last to first.

4. Search Patient – Find a patient using Patient ID.

5. Delete Patient – Remove a selected patient record.

6. Exit – Terminate the program safely.